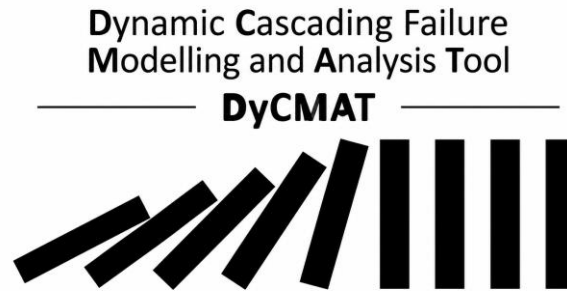


DyCMAT Tool Manual



Dynamic Cascading Failure Modelling and Analysis Tool

Version: v2.1

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1. Overview

DyCMAT is a research tool that interfaces with DIgSILENT PowerFactory through its Python API to support dynamic cascading-failure studies. It automates model initialization, RMS/dynamic simulation, protection-model installation, contingency generation or import, batch cascading simulations, and post-processing of cascading results. The tool is intended for power-system researchers and engineers performing dynamic security, resilience, and cascading-impact studies.

2. Main Functionalities

- Initialization and RMS dynamic simulation automation.
- Random N-k cascading contingency analysis.
- Automatic outage-event generation for transmission lines.
- Dynamic cascading modelling workflows.
- Automated protection and relay modelling.
- Cascade-results processing and statistical analysis.
- Quantifying cascading impacts
- Identifying power system vulnerability to cascading failures

2. How to Use

- 1- Download the .exe file from the link below:

[SnHashemi86/Dynamic-Cascading-Failure-Modelling-and-Analysis-Tool-DyCMAT-: Dynamic Cascading Failure Modelling and Analysis Tool Using the DIgSILENT PowerFactory and Python API.](https://github.com/SnHashemi86/Dynamic-Cascading-Failure-Modelling-and-Analysis-Tool-DyCMAT-)

- 2- Open or prepare an existing dynamic power-system model in DIgSILENT PowerFactory and ensure the required study case is available.

- 3- Launch DyCMAT and specify the PowerFactory Python API path, project name, and optional study-case name.
- 4- Initialize the system and, if required, run a test RMS/dynamic simulation to verify the model and retrieve the system inventory.
- 5- Select the required cascading-protection functions and configure the relay thresholds, delays, and shedding settings; then install/update the selected models.
- 6- Define the initiating contingencies either by generating random N-k line sets or by importing user-defined sets from an Excel file.
- 7- Set outage timing, simulation duration, and the output folder, then run the cascading analysis.
- 8- Open the Results & Visualization tab, select the result folder, process the saved result files, and review/export metrics statistics, component contributions, and plots.

3. System Requirements

- Microsoft Windows 10 or newer.
- DIgSILENT PowerFactory 2021 or earlier versions.
- Valid DIgSILENT PowerFactory license.
- PowerFactory Python 3.10 API environment.
- Available PowerFactory projects and study cases.

4. Setup

1. Check the DIgSILENT PowerFactory to be installed.
2. Launch DyCMAT.exe.
3. PowerFactory Python API path: default C:/Program Files/DIgSILENT/PowerFactory 2023 SP2/Python/3.10.
4. PowerFactory project name: default “39 Bus New England System”; study-case name is optional (blank uses the currently active study case).
5. Protection settings, including thresholds and delays for UFLS/UVLS stages (frequency threshold, delay, and shed percentage), generator over-/under-frequency protection, generator/source over-/under-voltage protection, and overload protection.
6. Contingency source: random N-k generation or Excel (.xlsx).
7. Outage timing: first outage time and time step between successive outages within a contingency set.
8. Simulation settings: simulation duration.
9. Output location: user-selectable folder for PowerFactory log/result files; processed CSV tables and PNG/JPG plots are exported to the selected results folder.

Note: These configurations are set by default; however, users are advised to check them beforehand. If the model has already been set up and run at least once, there is no need to repeat the cascade modelling. Users can proceed directly to the cascading analysis and process the results.

5. Tabs Overview

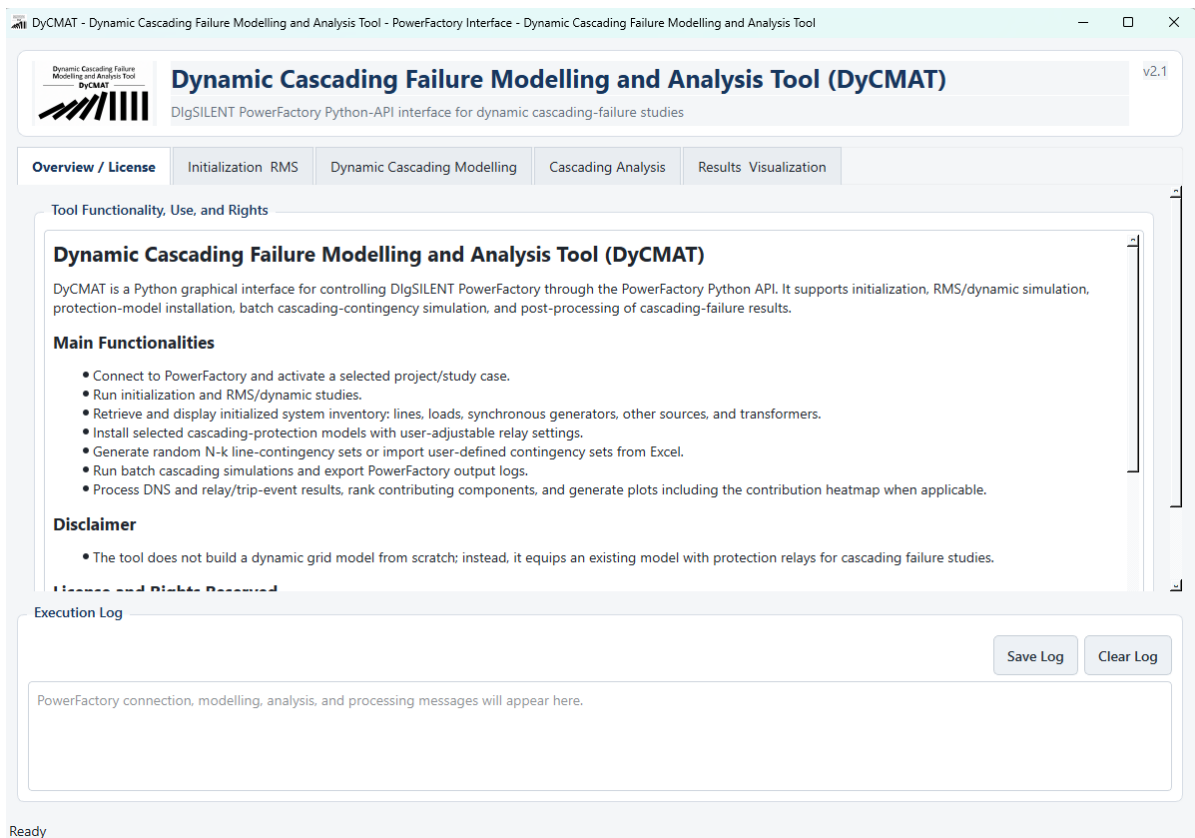
Overview / License: Provides the tool description, intended use, disclaimer, license and rights information, system requirements, and related references.

Initialization / RMS: Used to connect to DlgSILENT PowerFactory, activate the selected project/study case, initialize the system, retrieve system information, and execute RMS/dynamic simulations.

Dynamic Cascading Modelling: Used to install and configure the protection and relay models required for dynamic cascading-failure studies.

Cascading Analysis: Used to define, generate, and execute random or user-defined N-k cascading contingencies and export the corresponding PowerFactory simulation outputs.

Results Visualization: Used to process exported cascading-simulation results, quantify cascade metrics, including DNS and component-contribution metrics, and generate result tables, plots, and visualizations.



6. Disclaimer

This software is intended for academic, research, and engineering purposes only. Users are responsible for independently validating the underlying models, simulation settings, and all results obtained using the tool.

7. Related Publications

S. Hashemi, M. Asprou, L. Hadjidemetriou and M. Panteli, "Quantifying and Mitigating Cascading Impacts in HVdc-Interconnected Power Grids," IEEE Access, vol. 13, pp. 154491-154507, 2025.

S. Hashemi, V. S. Rajkumar, A. Ştefanov, and M. Panteli, "Cyber-physical-aware cascading mitigation in converter-dominated power systems," Sustainable Energy, Grids and Networks, vol. 44, p. 102065, 2025.